



# EXPERT REPORT

Assessment of Social Engineering Cybercrime in India

## **Risk Assessment, Legal Effectiveness, Digital Forensic Investigation and Policy Recommendations**

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## EXECUTIVE SUMMARY

Social engineering has become a significant cyber threat in India as rapid digital transformation and increased reliance on online financial and public services have expanded opportunities for cyber-enabled fraud. By exploiting human psychology rather than technical vulnerabilities, offenders employ techniques such as phishing, impersonation, digital arrest scams and Business Email Compromise (**BEC**) to obtain sensitive information and commit financial offences.

This report evaluates social engineering cybercrime in India, examining its ethical, legal and investigative implications. It assesses the effectiveness of the current legislative framework, digital forensic capability and investigative practices, concluding that although existing legislation provides a strong basis for prosecution, its effectiveness is constrained by evolving attack techniques, transnational criminal activity and the growing use of artificial intelligence.

The report emphasises the importance of reliable digital evidence, ethical investigative practices and effective digital forensic procedures. It recommends an integrated national strategy centred on legislative reform, strengthened digital forensic capability, continuous public awareness, enhanced public-private and international cooperation, and the responsible use of artificial intelligence to strengthen India's long-term cyber resilience.

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# 1. Introduction

Social engineering is one of the most significant cyber threats affecting governments, organisations and individuals because it exploits human psychology rather than technical vulnerabilities. Rather than bypassing security controls, offenders manipulate trust, authority, fear and urgency to persuade victims to disclose confidential information, transfer funds or compromise sensitive systems (Hahnagy, 2018; Holt, Bossler and Seigfried-Spellar, 2022). As organisations strengthen technical defences through encryption and multi-factor authentication, attackers increasingly target people as the weakest element of cybersecurity.

India's rapid digital transformation has expanded opportunities for social engineering attacks. The widespread adoption of Unified Payments Interface (**UPI**), digital banking, e-commerce and online public services has increased exposure to phishing, impersonation, digital arrest scams and identity theft. Recent CERT-In statistics indicate a continuing rise in reported cyber incidents, highlighting the growing scale of cyber-enabled threats across the country (CERT-In, 2026).

This expert report evaluates social engineering cybercrime in India for a government policy advisory agency. It critically examines the ethical and legal issues associated with these offences, evaluates the effectiveness of the current legislative and investigative framework, assesses the role of digital forensic evidence, considers the impact on victims and society, and proposes policy recommendations to strengthen national cyber resilience.

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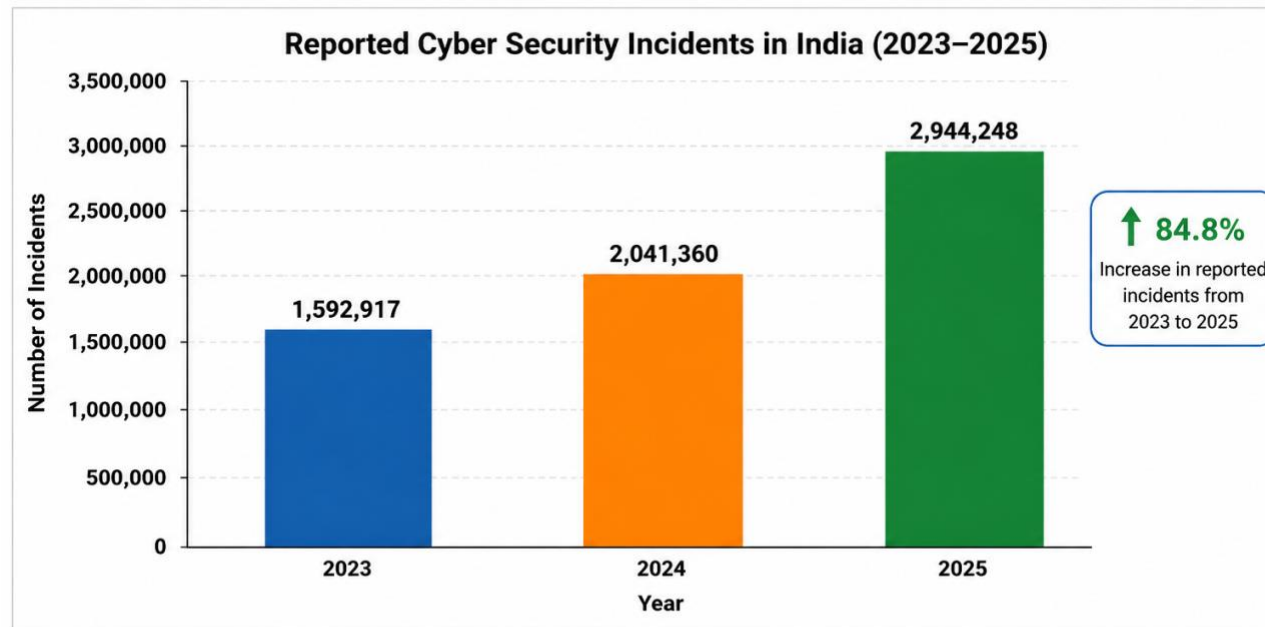
## 2. Social Engineering Cybercrime in India

### 2.1 Manifestation and Current Threat Landscape

Social engineering has become one of the fastest-growing forms of cybercrime in India because it exploits human behaviour rather than technical vulnerabilities. Rapid digital transformation, driven by widespread adoption of Unified Payments Interface (**UPI**), internet banking, e-commerce and digital government services, has expanded opportunities for offenders to manipulate individuals through deception instead of directly attacking computer systems (Holt, Bossler and Seigfried-Spellar, 2022).

Government data demonstrates the increasing scale of cyber-enabled crime. According to the Indian Computer Emergency Response Team (**CERT-In**), reported cyber security incidents increased from **1,592,917** in 2023 to **2,041,360** in 2024 and **2,944,248** in 2025 (CERT-In, 2026). Although these figures cover multiple forms of cybercrime, phishing, impersonation and financial fraud remain among the most prevalent threats affecting Indian citizens.

**Figure 1. Reported Cyber Security Incidents in India (2023–2025)**  
(Source: Indian Computer Emergency Response Team (CERT-In), Annual Report 2025–26)



**Figure 1** illustrates a sustained increase in reported cyber incidents, highlighting that technical security improvements alone have not reduced the overall threat. This reinforces the need for stronger public awareness, effective legislation and coordinated law enforcement.

## 2.2 Major Social Engineering Techniques and Case Examples

Social engineering in India has evolved from conventional phishing into sophisticated, psychologically driven attacks that exploit public trust in financial institutions, government agencies and digital platforms. Rather than targeting technical vulnerabilities, offenders manipulate victims into voluntarily disclosing sensitive information or authorising fraudulent transactions.

Phishing remains one of the most common attack methods. Criminals use fraudulent emails, SMS messages and fake websites impersonating banks, e-commerce platforms and government departments to obtain banking credentials, one-time passwords (OTPs) and personal information (Holt, Bossler and Seigfried-Spellar, 2022).

Digital arrest scams have emerged as one of India's most serious social engineering threats. Offenders impersonate police officers, Central Bureau of Investigation (CBI) officials or other government authorities, falsely accusing victims of offences such as money laundering or identity theft before coercing them into transferring money during prolonged video calls. The Ministry of Home Affairs has repeatedly advised that legitimate law enforcement agencies neither conduct investigations nor request financial transfers through video conferencing platforms (Ministry of Home Affairs, 2025).

Business Email Compromise (**BEC**) also presents a significant organisational threat. Attackers impersonate senior executives or trusted suppliers to induce employees to authorise fraudulent payments or disclose confidential corporate information, exploiting organisational trust rather than technical weaknesses (Verizon, 2025).

Despite differing delivery methods, these attacks consistently exploit authority, urgency, fear and trust. Consequently, effective mitigation requires public awareness, robust verification procedures and timely law enforcement alongside technical security controls.

## Figure 2. Typical Social Engineering Attack Lifecycle

(Adapted from Hadnagy (2018) and Holt, Bossler and Seigfried-Spellar (2022))



Figure 2 illustrates the typical lifecycle of a social engineering attack. Although techniques vary, most attacks follow a similar progression from reconnaissance and impersonation to psychological manipulation, resulting in disclosure of sensitive information or fraudulent actions.

## 2.3 Factors Contributing to the Growth of Social Engineering in India

The growth of social engineering in India reflects the interaction between rapid digital adoption, uneven cybersecurity awareness and increasingly sophisticated criminal techniques. Rather than exploiting software vulnerabilities, offenders target human decision-making, allowing attacks to remain effective despite advances in technical security.

A major contributing factor is the widespread adoption of digital payment platforms, particularly the Unified Payments Interface (**UPI**). Although UPI has transformed digital transactions through speed and convenience, it has also expanded opportunities for phishing, impersonation and payment fraud. Offenders exploit user's familiarity with digital payment requests by creating a false sense of urgency or authority, encouraging them to authorise transactions without independently verifying their legitimacy (Reserve Bank of India, 2024).

Cybersecurity awareness also remains inconsistent across India's diverse population. Despite awareness campaigns by government agencies and financial institutions, many individuals remain vulnerable to QR code fraud, remote access scams and AI-assisted impersonation. Criminals rapidly adapt their techniques, reducing the long-term effectiveness of awareness campaigns alone.

Artificial intelligence has further strengthened social engineering by enabling criminals to generate convincing phishing emails, synthetic voices and deepfake videos at scale, reducing both cost and technical barriers to entry (Choraś and Woźniak, 2021). Consequently, social engineering should be recognised as a socio-technical challenge requiring coordinated legal, technical and educational responses.

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## 3. Rights and Ethical Considerations

### 3.1 Balancing Cybersecurity, Privacy and Individual Rights

The investigation and prevention of social engineering offences require governments to balance effective law enforcement with the protection of fundamental rights. Authorities need sufficient powers to investigate cybercrime and obtain digital evidence, but these powers must remain proportionate, transparent and subject to legal oversight. Excessive surveillance or intrusive investigative practices risk undermining privacy, freedom of expression and public confidence in digital governance (Montasari and Hill, 2019).

In India, investigators obtain digital evidence from financial institutions, telecommunications providers and online platforms to identify offenders involved in phishing, impersonation and digital payment fraud. Although such access is often necessary, the collection and processing of personal data must comply with safeguards that prevent unnecessary interference with individual rights.

The Digital Personal Data Protection Act 2023 (**DPDP Act**) establishes principles for the lawful processing of personal data while recognising limited exemptions for law enforcement and national security (Government of India, 2023b). These exemptions highlight the continuing challenge of enabling effective cybercrime investigations while preserving privacy, accountability and the rule of law.

### 3.2 Ethical Responsibilities in Digital Forensics and Artificial Intelligence

Ethical practice is fundamental to digital forensic investigations because the reliability of digital evidence depends on both technical competence and professional integrity. Practitioners must ensure that evidence is identified, collected, preserved, analysed and reported objectively while maintaining an unbroken chain of custody. Ethical failures at any stage may compromise evidential integrity and weaken criminal proceedings (Holt, Bossler and Seigfried-Spellar, 2022).

Professional standards reinforce these responsibilities. The BCS Code of Conduct and the IEEE Code of Ethics require practitioners to demonstrate honesty, competence, accountability and respect for confidentiality and the public interest (BCS, 2021; IEEE, 2021). These principles are particularly important when investigations involve sensitive personal or financial information.

Artificial intelligence is increasingly used to detect phishing campaigns, analyse digital evidence and identify suspicious behaviour. Although AI improves investigative efficiency, it may generate false positives or misinterpret contextual factors such as language, culture or intent. Consequently, AI should support rather than replace professional judgement, with meaningful human oversight ensuring fairness, accountability and evidential reliability (Choraś and Woźniak, 2021).

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## 4. Critical Evaluation of India's Legal Framework

### 4.1 Legislative Framework Governing Social Engineering Cybercrime

India does not criminalise social engineering as a distinct offence. Instead, offenders are prosecuted under the Information Technology Act 2000 (as amended) and the Bharatiya Nyaya Sanhita (**BNS**) 2023, depending on the nature of the offence. Together, these statutes address identity theft, cheating by personation, unauthorised access to computer systems and other forms of cyber-enabled fraud (Government of India, 2000; Government of India, 2023a).

The Information Technology Act remains India's principal cybercrime legislation, while the Bharatiya Nyaya Sanhita modernises broader criminal law relating to fraud and deception. This framework enables investigators to prosecute offences arising from phishing, impersonation and digital payment fraud. However, because social engineering is treated as a method of facilitating other offences rather than a standalone crime, investigations focus on the resulting fraud or identity theft rather than the deceptive conduct itself.



## 4.2 Effectiveness and Limitations of India's Legal Framework

India's legislative framework provides a broad legal basis for prosecuting offences arising from social engineering, particularly identity theft, financial fraud and unauthorised access to computer systems. By combining the Information Technology Act 2000 and the Bharatiya Nyaya Sanhita 2023, investigators can address a wide range of cyber-enabled offences without requiring separate legislation for each emerging attack technique (Government of India, 2000; Government of India, 2023a).

However, legislation evolves more slowly than cybercrime. The rapid emergence of artificial intelligence, deepfakes and synthetic identities requires investigators to apply legal provisions developed before many contemporary forms of digital deception. Although existing legislation remains broadly applicable, technological change continues to challenge its effectiveness.

The transnational nature of social engineering presents a further limitation. Organised criminal groups operate across jurisdictions, while differences in national legislation, evidential standards and mutual legal assistance procedures can delay investigations and reduce the likelihood of successful prosecution (UNODC, 2023).

Legislation alone cannot prevent social engineering because these offences exploit human behaviour rather than technical vulnerabilities. Long-term resilience therefore depends on public awareness, effective digital forensic investigation, organisational security and international cooperation supported by adaptable, technology-neutral legislation.

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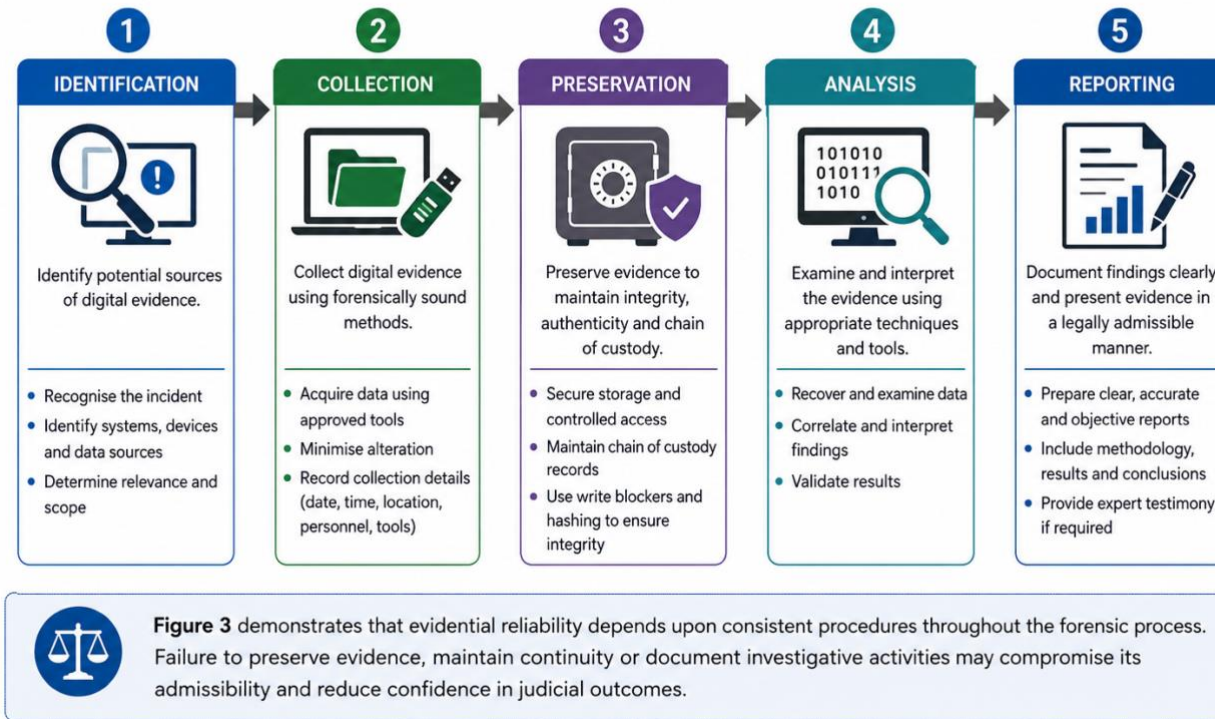
## 5. Critical Evaluation of Investigative Tools and Digital Forensic Capabilities

### 5.1 Digital Forensic Investigation of Social Engineering Offences

Effective investigation of social engineering offences depends on the timely identification, preservation and analysis of digital evidence. Unlike conventional crime, these investigations involve multiple evidence sources, including emails, mobile devices, messaging platforms, banking records, IP logs, CCTV footage and cloud services. Maintaining forensic integrity throughout the evidence lifecycle is therefore essential to ensure evidential reliability (Casey, 2020).

Digital investigations should follow recognised procedures for identifying, collecting, preserving and examining evidence. International standards such as ISO/IEC 27037 provide a framework for handling digital evidence while maintaining its authenticity and admissibility (ISO/IEC, 2012). An unbroken chain of custody remains essential to document evidence handling and preserve integrity throughout the investigation.

**Figure 3. Digital Forensic Investigation Process**  
*(Identification → Collection → Preservation → Analysis → Reporting)*



## 5.2 Evaluation of India's Investigative Capability

India has strengthened its cybercrime investigation capability through specialised organisations such as the Indian Cyber Crime Coordination Centre (**I4C**), the Indian Computer Emergency Response Team (**CERT-In**) and dedicated cybercrime police stations. These bodies provide technical expertise, incident response coordination and digital forensic support, improving the investigation of phishing, identity theft and other social engineering offences (Ministry of Home Affairs, 2025).

Despite these developments, significant operational challenges remain. Investigations frequently involve encrypted communications, cloud storage, cryptocurrency transactions and digital evidence located outside India's jurisdiction. International cooperation is often required, but differences in national legislation, evidential standards and mutual legal assistance procedures can delay investigations and reduce the likelihood of successful prosecution (UNODC, 2023).

Artificial intelligence offers valuable support by analysing large volumes of digital evidence, identifying phishing patterns and enhancing threat intelligence. However, AI should complement rather than replace professional judgement because automated systems may generate false positives or misinterpret contextual factors such as language, culture or intent (Choraś and Woźniak, 2021).

Effective investigations also depend on adherence to recognised forensic standards. Consistent evidence acquisition, preservation and documentation strengthen evidential reliability, while forensic reporting should retain sufficient professional flexibility to reflect the circumstances of each investigation. Continued investment in specialist expertise, international cooperation and the ethical use of AI will further strengthen India's investigative capability.

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## **6. Critical Evaluation of Victims, Harm, Perpetrators and Social Perception**

### **6.1 Victims and the Impact of Social Engineering**

Social engineering causes financial, psychological and social harm that extends beyond immediate financial loss. In addition to identity theft and fraud, victims may experience anxiety, embarrassment, reduced confidence and diminished trust in digital technologies after being manipulated into disclosing sensitive information or authorising fraudulent transactions. These impacts demonstrate that social engineering exploits human judgement rather than technical vulnerabilities (Holt, Bossler and Seigfried-Spellar, 2022).

Certain groups, including older adults, first-time digital banking users, students and individuals with limited cybersecurity awareness, remain particularly vulnerable. Digital arrest scams illustrate this by exploiting fear and perceived authority to coerce victims into transferring money or disclosing confidential information.

The impact also extends to organisations. Business Email Compromise (**BEC**) attacks can cause financial loss, operational disruption, disclosure of confidential information and reputational damage, while generating additional costs for incident response, digital forensic investigation and recovery (Verizon, 2025).

### **6.2 Perpetrators and Social Perception**

Modern social engineering is increasingly conducted by organised criminal networks operating across multiple jurisdictions. These groups exploit phishing infrastructure, credential theft, cryptocurrency laundering and AI-assisted impersonation to conduct large-scale fraud campaigns with limited technical expertise (Choraś and Woźniak, 2021).

Public perception remains a significant challenge because victims are often viewed as careless or technologically inexperienced, discouraging incident reporting and reinforcing stigma. However, social engineering exploits universal psychological characteristics such as trust, fear and obedience to authority, meaning that even experienced professionals can become victims. Reducing harm therefore requires stronger public awareness, institutional support for victims and coordinated action against organised criminal networks.

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## **7. Policy Recommendations**

This report concludes that legislation and technology alone cannot effectively reduce social engineering cybercrime. Strengthening India's cyber resilience requires coordinated action by government, law enforcement, financial institutions, private organisations and the public. The following recommendations are proposed.

### **7.1 Legislative Reform**

India's legal framework should be reviewed regularly to address emerging threats such as AI-generated impersonation, deepfake fraud and synthetic identities. Technology-neutral legislation will enable existing laws to remain effective as attack techniques evolve.

### **7.2 Strengthen Digital Forensic Capability**

Continued investment in digital forensic laboratories, specialist training and ISO/IEC 27037-compliant procedures should remain a national priority to strengthen the integrity, reliability and admissibility of digital evidence.

### **7.3 Expand Public Awareness and Cyber Education**

Continuous cyber awareness programmes should provide practical guidance on recognising phishing, impersonation, QR code fraud and digital arrest scams while encouraging verification of digital communications without blaming victims.

### **7.4 Strengthen Public-Private and International Cooperation**

Closer collaboration between law enforcement, financial institutions, technology companies and international partners should improve information sharing, cross-border evidence collection and the investigation of organised cybercrime.

### **7.5 Promote Ethical Use of Artificial Intelligence**

Artificial intelligence should support fraud detection and digital forensic investigations while remaining subject to meaningful human oversight, transparency and accountability.

**Table 1. Summary of Policy Recommendations**

| Policy Area                 | Key Recommendation                                                                                                                       | Expected Outcome                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Legislative Reform          | Regularly review legislation to address AI-enabled deception and emerging social engineering techniques through technology-neutral laws. | A flexible legal framework capable of responding to evolving cyber threats.           |
| Digital Forensic Capability | Expand specialist laboratories, investigator training and ISO/IEC 27037-compliant forensic procedures across investigative agencies.     | Improved evidential integrity, investigative consistency and successful prosecutions. |
| Public Awareness            | Deliver continuous cyber awareness and education programmes across schools, universities, workplaces and communities.                    | Increased public resilience and reduced victimisation.                                |
| International Cooperation   | Strengthen cross-border evidence sharing, intelligence exchange and operational collaboration between public and private sectors.        | Faster investigations and improved disruption of organised cybercrime networks.       |
| Artificial Intelligence     | Ensure AI supports investigations through transparent, accountable and human-supervised decision-making.                                 | Ethical, reliable and trustworthy AI-assisted investigations.                         |

**Source:** Compiled from Holt, Bossler and Seigfried-Spellar (2022), ISO/IEC (2012), UNODC (2023), Ministry of Home Affairs (2025) and the analysis presented in this report.

Collectively, these recommendations provide a practical framework for strengthening India’s resilience against social engineering attacks through legislative reform, digital forensic capability, ethical governance, technological innovation and sustained public awareness.

## 8. Conclusion

Social engineering remains one of India's most significant cyber threats because it exploits human behaviour rather than technical vulnerabilities. While rapid digital transformation has delivered substantial economic and social benefits, it has also increased opportunities for phishing, impersonation, digital arrest scams and other forms of cyber-enabled fraud.

This report has demonstrated that India possesses a comprehensive legal and institutional framework for investigating and prosecuting social engineering offences. However, its effectiveness is constrained by evolving attack techniques, transnational criminal networks and the growing use of artificial intelligence. Consequently, legislation alone cannot provide an effective response without robust digital forensic capability, international cooperation and sustained public awareness.

Strengthening India's resilience therefore requires an integrated national strategy combining effective legislation, reliable digital forensic investigation, ethical governance, public awareness and collaboration between government, industry and international partners. Implementing these recommendations will improve the prevention, investigation and prosecution of social engineering cybercrime while supporting long-term national cyber resilience.

# References

BCS, The Chartered Institute for IT (2021) *BCS Code of Conduct*. Swindon: BCS.

Casey, E. (2020) *Digital Evidence and Computer Crime: Forensic Science, Computers and the Internet*. 4th edn. London: Academic Press.

CERT-In (2026) *Annual Report 2025-26*. New Delhi: Ministry of Electronics and Information Technology. Available at: <https://www.cert-in.org.in/> (Accessed: 17 July 2026).

Choraś, M. and Woźniak, M. (2021) 'The double-edged sword of AI: Ethical adversarial attacks to counter artificial intelligence for crime', *AI and Ethics*, 2(4), pp. 631-634.

Government of India (2000) *Information Technology Act, 2000*. New Delhi: Government of India.

Government of India (2023a) *Bharatiya Nyaya Sanhita, 2023*. New Delhi: Government of India.

Government of India (2023b) *Digital Personal Data Protection Act, 2023*. New Delhi: Government of India.

Hadnagy, C. (2018) *Social Engineering: The Science of Human Hacking*. 2nd edn. Indianapolis: Wiley.

Holt, T.J., Bossler, A.M. and Seigfried-Spellar, K.C. (2022) *Cybercrime and Digital Forensics*. 3rd edn. New York: Routledge.

IEEE (2021) *IEEE Code of Ethics*. Piscataway, NJ: Institute of Electrical and Electronics Engineers.

ISO/IEC (2012) *ISO/IEC 27037:2012 Guidelines for Identification, Collection, Acquisition and Preservation of Digital Evidence*. Geneva: International Organization for Standardization.

Ministry of Home Affairs (2025) *Indian Cyber Crime Coordination Centre (I4C): Public Advisories on Digital Arrest Scams*. Government of India. Available at: <https://www.cybercrime.gov.in/> (Accessed: 17 July 2026).

Montasari, R. and Hill, R. (2019) 'Next-Generation Digital Forensics: Challenges and Future Paradigms', *2019 IEEE 12th International Conference on Global Security, Safety and Sustainability*, pp. 1-6.

Reserve Bank of India (2024) *Annual Report 2023-24*. Mumbai: Reserve Bank of India. Available at: <https://www.rbi.org.in/> (Accessed: 17 July 2026).

United Nations Office on Drugs and Crime (UNODC) (2023) *Comprehensive Study on Cybercrime*. Vienna: United Nations. Available at: <https://www.unodc.org/> (Accessed: 17 July 2026).

Verizon (2025) *2025 Data Breach Investigations Report*. New York: Verizon Business. Available at: <https://www.verizon.com/business/resources/reports/dbir/> (Accessed: 17 July 2026).